estimators $\hat{\alpha}$ and $\hat{\beta}$ are the unbiased estimators of the unknown population parameters α and β.

Here the regression coefficient β measures the rate of change of Y per unit change of X.

Example 9.2

In this example we shall consider the issue of fitting a Keynesian consumption function to the Indian data. Table 9.2 presents the GDP at factor cost and final consumption expenditure (FCE) figures for the Indian economy during the period 1980-2001 at 1993-94 prices.

Table 9.2: GDP and Final Consumption Expenditure (Rs. Crore)

Year	GDP	FCE	Year	GDP	FCE
1980	401128	390671	1991	701863	620806
1981	425073	407728	1992	737792	637307
1982	438079	415924	1993	781345	666950
1983	471742	446396	1994	838031	695825
1984	492077	461675	1995	899563	740109
1985	513990	485090	1996	970082	794093
1986	536257	504854	1997	1016595	826834
1987	556778	525558	1998	1082747	888513
1988	615098	557527	1999	1148367	952489
1989	656331	584970	2000	1198592	972932
1990	692871	610169	2001	1267945	1027254

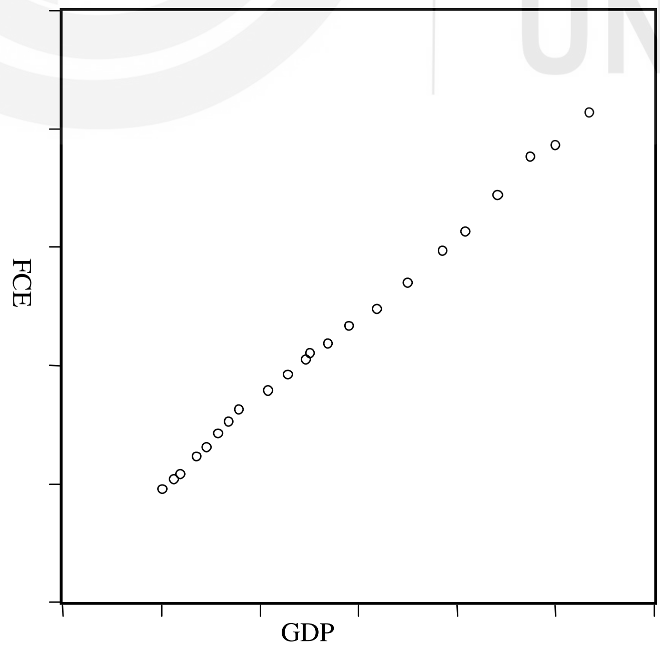

Fig. 9.3: Scatter of Final Consumption Expenditure against GDP

The above scatter of final consumption expenditure against GDP been obtained from Table 9.2. The scatter makes it clear that there seems to be a linear relationship between the two variables. Consequently, we have run a linear regression between the variables by using the data of Table 9.2. The results are presented below:

$$FCE = 108206.4 + 0.719674\ GDP$$

$$s.e.(\hat{\alpha}) = 6233.203 \qquad s.e.(\hat{\beta}) = 0.007865$$

$$t\ (\hat{\alpha}) = 17.35968 \qquad t\ (\hat{\beta}) = 91.50314$$

$$\mathrm{R}^2 = 0.997617 \qquad d.f. = 20$$

A high R^2 of more than 0.99 implies a tight fit. The estimated $\hat{\beta} = 0.719674$ indicates a marginal propensity to consume of about 72 percent and this can be quite expected in the Indian economy. We shall explain the significance of degrees of freedom and *t* values later in the issue of tests of hypotheses. However, we shall continue to present these two statistics in our subsequent examples also.

2) **Log-linear Model**

This model is also known as log-log, double-log or constant elasticity model. Its original form is given by

$Y = \alpha X^{\beta} e^{U}$, Where, *e* is the base of natural log (*e* is approximately equal to 2.718). In the original form, the model is non linear in nature. However, by applying the log transformation, the model is transformed to

$$\ln Y = \ln \alpha + \beta \ln X + U$$

If we put $\ln Y = y$, $\ln \alpha = a$, $\beta = b$ and $\ln X = x$, the model reduces to

$$y = a + bx + U$$

From the above transformation, it is very clear that the model now becomes linear in parameter and linear in log of the variables; and thus, can be estimated by the ordinary least square method. Here, *a* will be estimating $\ln \alpha$ and then by taking the antilog of *a*, we shall obtain the estimate for α . The regression coefficient $b = \beta$ deserves our special attention. It measures a change in log *Y* per unit of a change in log *X*. In the language of calculus, $\beta = \frac{d \ln Y}{d \ln X}$. Now, a change in the log of some variable implies a proportional change in it. Thus, β is the ratio of a proportional change in *Y* to a proportional change in *X*. in other words, β measures the elasticity of *Y* with respect to *X*. This is the rationale for the log linear regression model being termed as the constant elasticity model.

Example 9.3

This is an example from Gujrati (2003). The table below presents the quarterly data on total personal consumption expenditure (PCExp) and expenditure on durables (ExpDur) for the U.S. economy measured in 1992 billions of dollar for the period 1993:1-1998:3.

Table 9.3: Personal Consumption Expenditure and Expenditure on Durables in U.S.A.

Year	PCExp	ExpDur
1993:1	4286.8	504
1993:2	4322.8	519.3
1993:3	4366.6	529.9
1993:4	4398	542.1
1994:1	4439.4	550.7
1994:2	4472.2	558.8
1994:3	4498.2	561.7
1994:4	4534.1	576.6
1995:1	4555.3	575.2
1995:2	4593.6	583.5
1995:3	4623.4	595.3
1995:4	4650	602.4
1996:1	4692.1	611
1996:2	4746.6	629.5
1996:3	4768.3	626.5
1996:4	4802.6	637.5
1997:1	4853.4	656.3
1997:2	4872.7	653.8
1997:3	4947	679.6
1997:4	4981	648.8
1998:1	5055.1	710.3
1998:2	5130.2	729.4
1998:3	5181.8	733.7

Source: *Damodar Gujrati (2003), Basic Econometrics*

We have plotted the log of expenditure on durables against the log of total personal consumption expenditure in Figure 9.4 below.

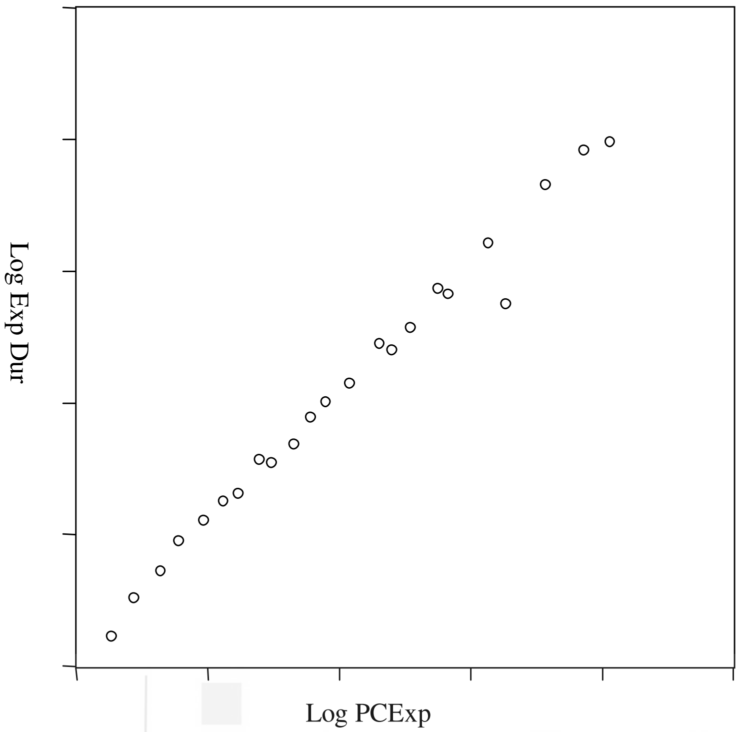

Fig. 9.4: Scatter of Log of Exp on Durables against Log of Personal Con Exp

The scatter above is clearly indicative of a linear relationship between the log of the two variables. As a result, we fit in a log linear model to the data on total personal consumption expenditure and the expenditure on consumer durables. We present some of the results from the fit below.

$$\ln ExpDur = -9.697098 + 1.905633 \ \ln PCExp$$

$$s.e.(\ln \hat{\alpha}) = 0.434127 \qquad s.e.(\hat{\beta}) = 0.051370$$

$$t \ (\ln \hat{\alpha}) = -22.33702 \qquad t \ (\hat{\beta}) = 37.09622$$

$$R^2 = 0.984969 \qquad d.f. = 21$$

We have an R^2 of about 0.99. This is indicative of a good fit. In this fit, the slope coefficient is an estimate of the elasticity of expenditure on durables with respect to personal consumption expenditure. Thus, $\hat{\beta} = 1.905633$ indicates that the expenditure on durables is highly elastic to total personal consumption expenditure.

3) **Semi-Log Model**

One of the common forms of semi-log regression models is the so-called growth rate model. The model is expressed as

$$\log Y = \alpha + \beta t + U$$

In this model the dependent variable is expressed in log, whereas, the independent variable is the time period, and it is measured in absolute term. Since, log operator is applied only to the left hand side of the equation, the model is called the semi-log model. In this model, the slope coefficient β is a measure of the proportional change in the dependent variable for a unit change in the time period. Thus it measures the growth rate of the dependent variable.

Example 9.4

In this example, we are going to verify the so-called 'Hindu Rate of Growth' of about 3.5 per cent that Indian economy consistently witnessed in 1960s and 1970s.

In Table 9.4, we present India's GDP at 1993-94 prices for the period 1960-1980.

Table 9.4: India's GDP during 1970-1980

Year	GDP	Year	GDP
1960	206103	1971	299269
1961	212499	1972	298316
1962	216994	1973	311894
1963	227980	1974	315514
1964	245270	1975	343924
1965	236306	1976	348223
1966	238710	1977	374235
1967	258137	1978	394828
1968	264873	1979	374291
1969	282134	1980	401128
1970	296278		

Figure 9.5 presents the scatter of the log of GDP against the time period denoted by the letter *t*.

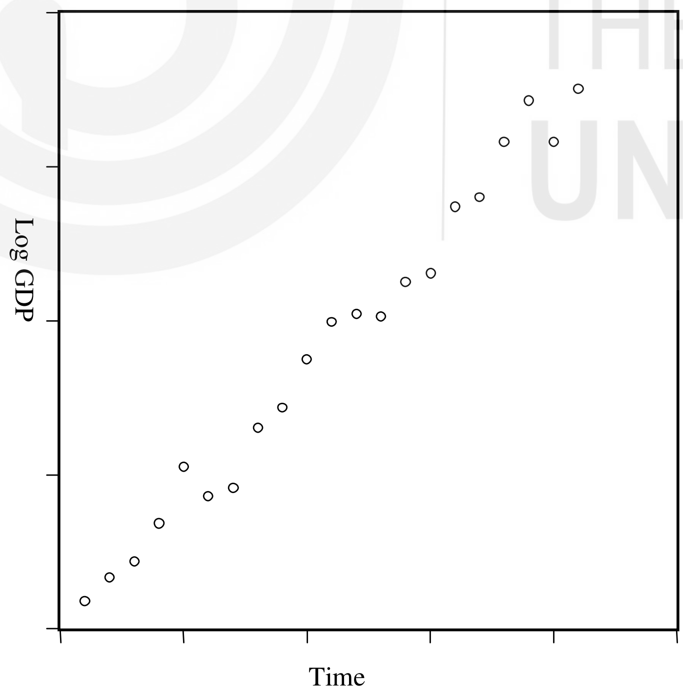

Fig. 9.5: Scatter of India's GDP against Time

The scatter above suggests a linear relationship between the log of GDP and the time period. Therefore, we run a regression of log GDP on time period *t*. we present the result below.

$$\log GDP = 12.19473 + 0.033729\,t$$

$$s.e.(\hat{\alpha}) = 0.012517 \qquad s.e.(\hat{\beta}) = 0.000997$$

$$t(\hat{\alpha}) = 974.2658 \qquad t(\hat{\beta}) = 33.83607$$

$$R^2 = 0.983675 \qquad d.f. = 19$$

The value of R^2 is quite high. The estimated slope coefficient of 0.034 indicates a rate of growth of 3.4 per cent during the period 1960-1980. This adequately supports the phenomenon of 'Hindu rate of Growth' in the 60s and the 70s.

4) **Reciprocal Model**

In the reciprocal model, the dependent variable is regressed on the reciprocal of the independent variable. Its functional from is

$$Y = \alpha + \beta \frac{1}{X} + U$$

Although the model is non-linear in variable (because the power of X is –1), it is linear in parameter. Hence, it can be estimated by the OLS method. The model has an important characteristic. As the independent variable X increases indefinitely, the term $\beta \frac{1}{X}$ approaches zero, β being a constant. Consequently, the dependent variable Y approaches a limiting value or an asymptotic value equal to the intercept α. An application of this model can be in the relationship between per capita GNP and the child mortality. As per capita GNP increases, the infant mortality rate is expected to fall but we cannot expect it to fall independently if per capita GNP continues to increase indefinitely. In such a situation, in all probability, the mortality rate will tend to a limiting value. Gujrati (1993) presents an example of applying the reciprocal model to study the relationship between the per capita GNP and the infant mortality rate by using the cross-section data for 64 countries. His results are presented below.

$$Y = 81.79436 + 27237.17 \frac{1}{X}$$

$$s.e.(\hat{\alpha}) = 10.8321 \qquad s.e.(\hat{\beta}) = 3759.999$$

$$t(\hat{\alpha}) = 7.5511 \qquad t(\hat{\beta}) = 7.2535$$

$$R^2 = 0.4590$$

where, Y is the child mortality rate (the number of deaths of children under the age of 5 per year per 1000 live births) and X is the per capita GNP at constant prices. The above-mentioned results show that if per capita GNP increases indefinitely, the infant mortality approaches a limiting rate of about 82 deaths per 1000 children born.

Another application of the reciprocal model can be in the estimation of the Phillips curve. This curve proposes an inverse relationship between the change in unemployment rate and that in inflation rate. With a growth in unemployment rate, the growth in inflation rate is expected to fall. In fact Phillips curve postulates that if the unemployment rate increases beyond its so-called natural rate, the growth in the inflation rate should become negative. However, with an indefinite increase in the unemployment rate, the inflation growth cannot be expected to fall indefinitely. It should stabilize at some negative limiting value. Thus, Phillips Curve can be an appropriate case for the use of the reciprocal model.

9.9 CLASSICAL NORMAL REGRESSION MODEL

We have already seen that a two variable classical linear regression model can be presented as

$$Y = \alpha + \beta X + U$$

In this model, U is the population disturbance term. We can estimate the unknown parameters of this regression model from the sample information by using the least square or, as it is sometimes called, ordinary least square method. The least square estimates possess some desirable properties if the population disturbance U satisfies some five assumptions that we have discussed in Section 9.3. These five assumptions are adequate for the estimation purposes. But the scope of a regression model is just not restricted to the estimation of the parameters. An important purpose of the sample estimates is to test some hypotheses about the unknown population regression parameters with their help. And we can do this if we make some assumption about the distribution of U. This we do by making the assumption of normality for U. The assumption essentially means that the population regression disturbance term follows normal distribution with mean zero, a constant variance equal to σ^2 and a zero covariance. In fact U has a conditional distribution, in the sense, that, for each of the given values of the non-stochastic independent variable X, we might have a distribution of different values of U. This will be clear if we reproduce the diagram of Figure 9.1 with the unknown population regression line fitted into it.

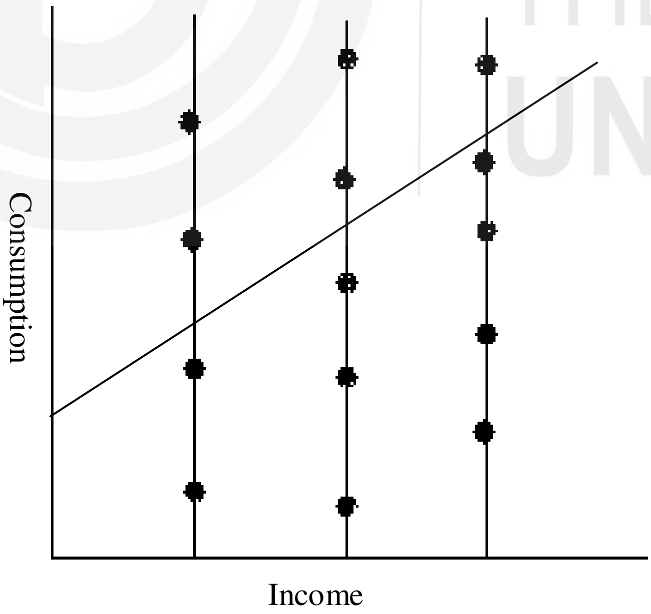

Fig. 9.6: Bi-Variate Population with the Unknown Regression Line

In Figure 9.6, we can see that for a given observed value of income, we can have different observed values of consumption represented by the bold dots on each of the vertical lines. The vertical distance between an observed value of consumption and the corresponding estimated value from the regression line for a given income level measures the value of the disturbance term. Thus there are many possible values of U for each of the given income levels. As a result, there are conditional distributions of U for different levels of income.

The significance of the normality assumption is that these conditional distributions of U should all be independently normally distributed with the same mean equal to zero and, the same variance equal to σ^2. Mathematically speaking,

$$E(U_i) = 0 \qquad \text{for all } i\text{s}$$
$$V(U_i) = \sigma^2 \qquad \text{for all } i\text{s}$$
$$Cov(U_i U_j) = 0 \quad \text{for } i \neq j$$

In other words, the population disturbance variable should have independent and identical normal distribution.

If we make this additional normality assumption along with the earlier-mentioned five assumptions about the disturbance term U, then the regression model is called the classical normal regression model.

It may be mentioned here that the normality assumption is important not only for hypothesis testing but also for the fact that under this assumption we can use an alternative procedure for estimating the population regression parameters. This procedure is known as the method of maximum likelihood estimation. Although for regression parameters the method gives the same estimates as the least square estimates, the approach followed is quite different. In fact this procedure has some stronger theoretical characteristics than the least square method. However, we are not discussing this procedure here because it is beyond our scope.

In the next section, we shall discuss the issue of hypothesis testing and see how the normality assumption plays a crucial role there.

9.10 HYPOTHESIS TESTING

Considering the two variable regression model

$$Y = \alpha + \beta X + U$$

We might be interested in examining whether the unknown parameter α or β assumes a particular value or not. This is what is known as hypothesis testing in statistics. We can conduct such a test from the sample estimate of α or β as the case might be. Although we may test some hypothesis about the intercept α, our main concern in regression model is the slope coefficient β. We have the estimated sample regression function

$$\hat{Y} = \hat{\alpha} + \hat{\beta} X + \hat{U}$$

Here, $\hat{\beta}$ is the sample estimate of β and we shall use it for estimating the unknown β. As mentioned earlier that $\hat{\beta}$ might vary from sample to sample collected from the same population. As a result, $\hat{\beta}$ is a random variable with some probability distribution. Now, for the purpose of hypothesis testing, we need to know the form of this distribution of $\hat{\beta}$.